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Daniel Kroening · Corina S. Păsăreanu (Eds.)

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Preface

It is our great pleasure to welcome you to CAV 2015, the 27th International Conference on Computer-Aided Verification, held in San Francisco, California, during July 18–24, 2015.

The CAV conference series is dedicated to the advancement of the theory and practice of computer-aided formal analysis methods for hardware and software systems. The conference covers the spectrum from theoretical results to concrete applications, with an emphasis on practical verification tools and the algorithms and techniques that are needed for their implementation. CAV considers it vital to continue spurring advances in hardware and software verification while expanding to new domains such as biological systems and computer security.

The CAV 2015 program included five keynotes, technical papers (58 long and 11 short papers accepted out of 252 submissions), 11 co-located events (VSTTE – Verified Software: Theories, Tools, and Experiments; SMT – Satisfiability Modulo Theories, EC2, IPRA – Interpolation: From Proofs to Applications; SYNT – Synthesis; VeriSure – Verification and Assurance; HCVS – Horn Clauses for Verification and Synthesis; VMW – Verification Mentoring Workshop, REORDER, SNR – Symbolic and Numerical Methods for Reachability Analysis; VEMDP – Verification of Engineered Molecular Devices and Programs), the Artifact Evaluation as well as briefings from the SMT and Synthesis competitions.

The invited keynote speakers were Philippa Gardner (Imperial College London), Leslie Lamport (Microsoft Research), Bob Kurshan (Cadence), William Hung (Synopsys), and Peter O’Hearn (University College London and Facebook).

Many people worked hard to make CAV 2015 a success. We thank the authors and the keynote speakers for providing the excellent technical material, the Program Committee for their thorough reviews and the time spent on evaluating all the submissions and discussing them during the on-line discussion period, and the Steering Committee for their guidance throughout the planning for CAV 2015.

We also thank Temesghen Kahsai, Local Chair, for his dedication and help with CAV 2015 planning and Hana Chockler, Sponsorship Chair, for helping to bring much needed financial support to the conference; Dirk Beyer, Workshop Chair, and all the organizers of the co-located events for bringing their events to the CAV week; Elizabeth Polgreen for the program and proceedings; Arie Gurfinkel, Temesghen Kahsai, Michael Tautschnig, and the Artifact Evaluation Committee for their work on evaluating the artifacts submitted.

We gratefully acknowledge NSF for providing financial support for student participants. We sincerely thank the CAV sponsors for their generous contributions:

- Google (Platinum sponsor)
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- Intel, Samsung (Bronze sponsors)

We also thank Carnegie Mellon University Silicon Valley and the University of Oxford for their support.

Finally, we hope you find the proceedings of CAV 2015 intellectually stimulating and practically valuable.

May 2015

Corina S. Păsăreanu
Daniel Kroening

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Piterman, Nir	Stefanescu, Gheorghe	Zeljić, Aleksandar
Podelski, Andreas	Stickse, Christoph	Zhang, Qirun
Pommellet, Adrien	Suda, Martin	Zhang, Yi
Pous, Damien	Sun, Chengnian	Zheng, Yunhui
Prasad, Mukul	Sun, Yutian	Zutshi, Aditya
Prähofer, Herbert	Szekeres, Laszlo	
Puggelli, Alberto		

Contents – Part II

SMT Techniques and Applications

POLING: SMT Aided Linearizability Proofs	3
<i>He Zhu, Gustavo Petri, and Suresh Jagannathan</i>	
Finding Bounded Path in Graph Using SMT for Automatic Clock Routing . . .	20
<i>Amit Erez and Alexander Nadel</i>	
Cutting the Mix	37
<i>Jürgen Christ and Jochen Hoenicke</i>	
The Inez Mathematical Programming Modulo Theories Framework	53
<i>Panagiotis Manolios, Jorge Pais, and Vasilis Papavasileiou</i>	
Using Minimal Correction Sets to More Efficiently Compute Minimal Unsatisfiable Sets	70
<i>Fahiem Bacchus and George Katsirelos</i>	
Deciding Local Theory Extensions via E-matching	87
<i>Kshitij Bansal, Andrew Reynolds, Tim King, Clark Barrett, and Thomas Wies</i>	

HW Verification

Modular Deductive Verification of Multiprocessor Hardware Designs	109
<i>Muralidaran Vijayaraghavan, Adam Chlipala, Arvind, and Nirav Dave</i>	
Word-Level Symbolic Trajectory Evaluation.	128
<i>Supratik Chakraborty, Zurab Khasidashvili, Carl-Johan H. Seger, Rajkumar Gajavelly, Tanmay Haldankar, Dinesh Chhatani, and Rakesh Mistry</i>	
Verifying Linearizability of Intel [®] Software Guard Extensions	144
<i>Rebekah Leslie-Hurd, Dror Caspi, and Matthew Fernandez</i>	

Synthesis

Synthesis Through Unification	163
<i>Rajeev Alur, Pavol Černý, and Arjun Radhakrishna</i>	

From Non-preemptive to Preemptive Scheduling Using Synchronization Synthesis.	180
<i>Pavol Černý, Edmund M. Clarke, Thomas A. Henzinger, Arjun Radhakrishna, Leonid Ryzhyk, Roopsha Samanta, and Thorsten Tarrach</i>	
Counterexample-Guided Quantifier Instantiation for Synthesis in SMT.	198
<i>Andrew Reynolds, Morgan Deters, Viktor Kuncak, Cesare Tinelli, and Clark Barrett</i>	
Deductive Program Repair	217
<i>Etienne Kneuss, Manos Koukoutos, and Viktor Kuncak</i>	
Quantifying Conformance Using the Skorokhod Metric	234
<i>Jyotirmoy V. Deshmukh, Rupak Majumdar, and Vinayak S. Prabhu</i>	
Pareto Curves of Multidimensional Mean-Payoff Games	251
<i>Romain Brenguier and Jean-François Raskin</i>	
Termination	
Conflict-Driven Conditional Termination	271
<i>Vijay D'Silva and Caterina Urban</i>	
Predicate Abstraction and CEGAR for Disproving Termination of Higher-Order Functional Programs	287
<i>Takuya Kuwahara, Ryosuke Sato, Hiroshi Unno, and Naoki Kobayashi</i>	
Complexity of Bradley-Manna-Sipma Lexicographic Ranking Functions	304
<i>Amir M. Ben-Amram and Samir Genaim</i>	
Measuring with Timed Patterns.	322
<i>Thomas Ferrère, Oded Maler, Dejan Ničković, and Dogan Ulus</i>	
Automatic Verification of Stability and Safety for Delay Differential Equations.	338
<i>Liang Zou, Martin Fränzle, Naijun Zhan, and Peter Nazier Mosaad</i>	
Time Robustness in MTL and Expressivity in Hybrid System Falsification. . .	356
<i>Takumi Akazaki and Ichiro Hasuo</i>	
Concurrency	
Adaptive Concretization for Parallel Program Synthesis	377
<i>Jinseong Jeon, Xiaokang Qiu, Armando Solar-Lezama, and Jeffrey S. Foster</i>	

Automatic Completion of Distributed Protocols with Symmetry	395
<i>Rajeev Alur, Mukund Raghothaman, Christos Stergiou, Stavros Tripakis, and Abhishek Udupa</i>	
An Axiomatic Specification for Sequential Memory Models	413
<i>William Mansky, Dmitri Garbuzov, and Steve Zdancewic</i>	
Approximate Synchrony: An Abstraction for Distributed Almost-Synchronous Systems	429
<i>Ankush Desai, Sanjit A. Seshia, Shaz Qadeer, David Broman, and John C. Eidson</i>	
Automated and Modular Refinement Reasoning for Concurrent Programs . . .	449
<i>Chris Hawblitzel, Erez Petrank, Shaz Qadeer, and Serdar Tasiran</i>	
Author Index	467

Contents – Part I

Invited Paper

A Trusted Mechanised Specification of JavaScript: One Year On	3
<i>Philippa Gardner, Gareth Smith, Conrad Watt, and Thomas Wood</i>	

Model Checking and Refinements

On Automation of CTL* Verification for Infinite-State Systems	13
<i>Byron Cook, Heidy Khlaaf, and Nir Piterman</i>	
Algorithms for Model Checking HyperLTL and HyperCTL*	30
<i>Bernd Finkbeiner, Markus N. Rabe, and César Sánchez</i>	
Fairness Modulo Theory: A New Approach to LTL Software Model Checking	49
<i>Daniel Dietsch, Matthias Heizmann, Vincent Langenfeld, and Andreas Podelski</i>	
Model Checking Parameterized Asynchronous Shared-Memory Systems	67
<i>Antoine Durand-Gasselin, Javier Esparza, Pierre Ganty, and Rupak Majumdar</i>	
SMT and POR Beat Counter Abstraction: Parameterized Model Checking of Threshold-Based Distributed Algorithms	85
<i>Igor Konnov, Helmut Veith, and Josef Widder</i>	
Skipping Refinement	103
<i>Mitesh Jain and Panagiotis Manolios</i>	

Quantitative Reasoning

Percentile Queries in Multi-dimensional Markov Decision Processes	123
<i>Mickael Randour, Jean-François Raskin, and Ocan Sankur</i>	
Faster Algorithms for Quantitative Verification in Constant Treewidth Graphs	140
<i>Krishnendu Chatterjee, Rasmus Ibsen-Jensen, and Andreas Pavlogiannis</i>	
Counterexample Explanation by Learning Small Strategies in Markov Decision Processes	158
<i>Tomáš Brázdil, Krishnendu Chatterjee, Martin Chmelík, Andreas Fellner, and Jan Křetínský</i>	

Symbolic Polytopes for Quantitative Interpolation and Verification	178
<i>Klaus von Gleissenthall, Boris Köpf, and Andrey Rybalchenko</i>	
Adaptive Aggregation of Markov Chains: Quantitative Analysis of Chemical Reaction Networks	195
<i>Alessandro Abate, Luboš Brim, Milan Češka, and Marta Kwiatkowska</i>	
PROPhESY: A PRObabilistic ParamETER SYNthesis Tool	214
<i>Christian Dehnert, Sebastian Junges, Nils Jansen, Florian Corzilius, Matthias Volk, Harold Bruintjes, Joost- Pieter Katoen, and Erika Ábrahám</i>	
Software Analysis	
Effective Search-Space Pruning for Solvers of String Equations, Regular Expressions and Length Constraints.	235
<i>Yunhui Zheng, Vijay Ganesh, Sanu Subramanian, Omer Tripp, Julian Dolby, and Xiangyu Zhang</i>	
Automata-Based Model Counting for String Constraints.	255
<i>Abdurbaki Aydın, Lucas Bang, and Tefvik Bultan</i>	
OpenJDK’s Java.util.Collection.sort() Is Broken: The Good, the Bad and the Worst Case	273
<i>Stijn de Gouw, Jurriaan Rot, Frank S. de Boer, Richard Bubel, and Reiner Hähnle</i>	
Tree Buffers	290
<i>Radu Grigore and Stefan Kiefer</i>	
Learning Commutativity Specifications	307
<i>Timon Gehr, Dimitar Dimitrov, and Martin Vechev</i>	
Angelic Verification: Precise Verification Modulo Unknowns	324
<i>Ankush Das, Shuvendu K. Lahiri, Akash Lal, and Yi Li</i>	
The SeaHorn Verification Framework	343
<i>Arie Gurfinkel, Temesghen Kahsai, Anvesh Komuravelli, and Jorge A. Navas</i>	
Automatic Rootcausing for Program Equivalence Failures in Binaries	362
<i>Shuvendu K. Lahiri, Rohit Sinha, and Chris Hawblitzel</i>	
Fine-Grained Caching of Verification Results	380
<i>K. Rustan M. Leino and Valentin Wüstholtz</i>	
Predicting a Correct Program in Programming by Example	398
<i>Rishabh Singh and Sumit Gulwani</i>	

Abstract Interpretation with Higher-Dimensional Ellipsoids and Conic Extrapolation.	415
<i>Mendes Oulamara and Arnaud J. Venet</i>	

Lightning Talks

ADAM: Causality-Based Synthesis of Distributed Systems	433
<i>Bernd Finkbeiner, Manuel Giesekeing, and Ernst-Rüdiger Olderog</i>	
Alchemist: Learning Guarded Affine Functions.	440
<i>Shambwaditya Saha, Pranav Garg, and P. Madhusudan</i>	
OptiMathSAT: A Tool for Optimization Modulo Theories	447
<i>Roberto Sebastiani and Patrick Trentin</i>	
Systematic Asynchrony Bug Exploration for Android Apps	455
<i>Burcu Kulahcioglu Ozkan, Michael Emmi, and Serdar Tasiran</i>	
Norn: An SMT Solver for String Constraints	462
<i>Parosh Aziz Abdulla, Mohamed Faouzi Atig, Yu-Fang Chen, Lukáš Holík, Ahmed Rezine, Philipp Rümmer, and Jari Stenman</i>	
PVSio-web 2.0: Joining PVS to HCI.	470
<i>Paolo Masci, Patrick Oladimeji, Yi Zhang, Paul Jones, Paul Curzon, and Harold Thimbleby</i>	
The Hanoi Omega-Automata Format	479
<i>Tomáš Babiak, František Blahoudek, Alexandre Duret-Lutz, Joachim Klein, Jan Křetínský, David Müller, David Parker, and Jan Strejček</i>	
The Open-Source LearnLib: A Framework for Active Automata Learning . . .	487
<i>Malte Isberner, Falk Howar, and Bernhard Steffen</i>	
BBS: A Phase-Bounded Model Checker for Asynchronous Programs	496
<i>Rupak Majumdar and Zilong Wang</i>	
Time-Aware Abstractions in HybridSal	504
<i>Ashish Tiwari</i>	
A Type-Directed Approach to Program Repair	511
<i>Alex Reinking and Ruzica Piskac</i>	
Formal Design and Safety Analysis of AIR6110 Wheel Brake System.	518
<i>M. Bozzano, A. Cimati, A. Fernandes Pires, D. Jones, G. Kimberly, T. Petri, R. Robinson, and S. Tonetta</i>	

Meeting a Powertrain Verification Challenge	536
<i>Parasara Sridhar Duggirala, Chuchu Fan, Sayan Mitra, and Mahesh Viswanathan</i>	
Synthesising Executable Gene Regulatory Networks from Single-Cell Gene Expression Data	544
<i>Jasmin Fisher, Ali Sinan Köksal, Nir Piterman, and Steven Woodhouse</i>	
Empirical Software Metrics for Benchmarking of Verification Tools	561
<i>Yulia Demyanova, Thomas Pani, Helmut Veith, and Florian Zuleger</i>	
Interpolation, IC3/PDR, and Invariants	
Property-Directed Inference of Universal Invariants or Proving Their Absence.	583
<i>A. Karbyshev, N. Bjørner, S. Itzhaky, N. Rinetzký, and S. Shoham</i>	
Efficient Anytime Techniques for Model-Based Safety Analysis	603
<i>Marco Bozzano, Alessandro Cimatti, Alberto Griggio, and Cristian Mattarei</i>	
Boosting k -Induction with Continuously-Refined Invariants	622
<i>Dirk Beyer, Matthias Dangl, and Philipp Wendler</i>	
Fast Interpolating BMC	641
<i>Yakir Vizel, Arie Gurfinkel, and Sharad Malik</i>	
Counterexample-Guided Polynomial Loop Invariant Generation by Lagrange Interpolation	658
<i>Yu-Fang Chen, Chih-Duo Hong, Bow-Yaw Wang, and Lijun Zhang</i>	
Author Index	675